



**University of the Punjab
(Gujranwala Campus)**

Lab Task

**Department of Information
Technology**

“Artificial Intelligence”

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TASK # 1

The screenshot shows the Programiz Python Online Compiler interface. The header includes the Programiz logo and the text "Python Online Compiler". Below the header, there is a tab labeled "main.py" and two buttons: "Visualize" and "Run". The code editor contains the following Python code:

```
1 x=1
2 #the initial value of x is 1
3 if x>0:
4     print("These are two comments")
5     #print a string
```

The output panel on the right displays the result of the code execution:

```
These are two comments

=== Code Execution Successful ===
```

TASK # 2

The screenshot shows the Programiz Python Online Compiler interface. The header includes the Programiz logo and the text "Python Online Compiler". Below the header, there is a tab labeled "main.py" and two buttons: "Visualize" and "Run". The code editor contains the following Python code:

```
1 print ("Statement1")
2 print("Statement2")
3 #you can write above two statements in the following way
4 print("Statement1"); print("Statement2")
5
```

The output panel on the right displays the result of the code execution:

```
Statement1
Statement2
Statement1
Statement2

=== Code Execution Successful ===
```

TASK # 3

Programiz

Python Online Compiler

Programiz PRO >

main.py

Visualize

Run ▶

```
1 x=1
2 if x>0:
3 print("this statement has no indentation")
4 print("This statement has no indentation")
5
```

Output

ERROR!

Traceback (most recent call last):

File "<main.py>", line 3

print("this statement has no indentation")

^^^^

IndentationError: expected an indented block after 'if' statement on line 2

=== Code Exited With Errors ===

TASK # 4

Programiz

Python Online Compiler

Program

main.py

Visualize

Run ▶

```
1 x=1
2 if x>0:
3 print("This statement has single space indentation")
4 print("This statement has single space indentation")
5 |
```

Output

This statement has single space indentation

This statement has single space indentation

=== Code Execution Successful ===

TASK # 5

Programiz

Python Online Compiler

P

main.py

Visualize

Run ▶

```
1 x=1
2 if x>0:
3 print("This statement has single tab indentation")
4 print("This statement has single tab indentation")
5
```

Output

This statement has single tab indentation

This statement has single tab indentation

=== Code Execution Successful ===

TASK # 6


Python Online Compiler

main.py

Visualize

Run ▶

```
1 x=1
2 if x>0:
3     print("This statement has single tab+space indentation")
4     print("This statement has single tab+space indentation")
5
```

Output

```
This statement has single tab+space indentation
This statement has single tab+space indentation

=== Code Execution Successful ===
```

Task 6

main.py

Visualize

Run ▶

```
1 a=123
2 print(type(a))
3
4 b=-4587
5 print(type(b))
6
7 c=0
8 print(type(c))
9
10 g=1.03
11 print (type(g))
12
13 h=-11.23
14 print(type(h))
15
16 i=.34
17 print(type(i))
18
19 j=2.12e-10
20 print(type(j))
21
22 k=5E220
23 print(type(k))
24
```

Output

```
<class 'int'>
<class 'int'>
<class 'int'>
<class 'float'>
<class 'float'>
<class 'float'>
<class 'float'>
<class 'float'>
<class 'float'>

=== Code Execution Successful ===
```

Task 7

main.py	Visualize	Run ▶	Output
<pre>1 x=complex(1,2) 2 print(type(x)) 3 print(x) 4 (1+2j) 5 z=1+2j 6 print(type(z)) 7 z=1+2J 8 print(type(z)) 9</pre>			<pre><class 'complex'> (1+2j) <class 'complex'> <class 'complex'> === Code Execution Successful ===</pre>

Task 8

main.py	Visualize	Run ▶	Output
<pre>1 x=True 2 print(type(x)) 3 y=False 4 print(type(y))</pre>			<pre><class 'bool'> <class 'bool'> === Code Execution Successful ===</pre>

Task 9

main.py	Visualize	Run ▶	Output
<pre>1 str1="string with double quotes" #string should be start and end with double quotes 2 print(str1) 3 str2='string with single quotes' 4 print(str2) 5 str3="string start with double and end with single quotes' 6 print(str3) 7 str4='string start with single and end with double quotes" 8 print(str4) 9 str5="Day's" #single quotes within double quotes 10 print(str5) 11 str6='Day"s' #double quotes within single quotes 12 print(str6)</pre>			<pre>ERROR! Traceback (most recent call last): File "<main.py>", line 5 str3="string start with double and end with single quotes' ^ SyntaxError: unterminated string literal (detected at line 5) === Code Exited With Errors ===</pre>

Task 10

main.py	Visualize	Run ▶	Output
<pre>1 print("This is a backslash (\\) mark.") 2 print("This is a tab \t key.") 3 print("These are \'single quotes\'") 4 print("These are \"double quotes\"") 5 print("This is a new line\nNew line.") 6</pre>			<pre>This is a backslash (\\) mark. This is a tab key. These are 'single quotes' These are "double quotes" This is a new line New line. === Code Execution Successful ===</pre>

Task 11

main.py	Visualize	Run ▶	Output
<pre>1 string1="PYTHON TUUTORIAL" 2 print(string1[0]) 3 print(string1[-15]) 4 print(string1[14]) 5 print(string1[-1]) 6 print(string1[4]) 7 print(string1[-11]) 8 print(string1[16]) # it gives error bcz 16 index doesn't exist</pre>			<pre>P Y A L O ERROR! N Traceback (most recent call last): File "<main.py>", line 8, in <module> IndexError: string index out of range === Code Exited With Errors ===</pre>

Task 12

Programiz Python Online Compiler			
main.py	Visualize	Run ▶	Output
<pre>1 my_list1=[5,12,13,14] #the list contains all integer values 2 print(my_list1) 3 my_list2=['red','blue','black','white'] # the list contain all string values 4 print(my_list2) 5 my_list3=['red',12,112.12] #this list contains integer, string and float values 6 print(my_list3)</pre>			<pre>[5, 12, 13, 14] ['red', 'blue', 'black', 'white'] ['red', 12, 112.12] === Code Execution Successful ===</pre>

Task 13

main.py	Visualize	Run ▶	Output
<pre>1 my_list=[] 2 print(my_list)</pre>			<pre>[] === Code Execution Successful ===</pre>

Task 14

main.py	Visualize	Run ▶	Output
<pre>1 color_list=["Red","Blue","Green","Black"] #te list contain 4 element indices start at 0 and end at 3 2 print(color_list[0]) #return the 1st element 3 print(color_list[1],color_list[3]) 4 print(color_list[-1]) 5 print(color_list[4]) #gives error bcz this index doesnot exist</pre>			<pre>Red Blue Black Black ERROR! Traceback (most recent call last): File "<main.py>", line 5, in <module> IndexError: list index out of range === Code Exited With Errors ===</pre>

Task 15

Programiz Python Online Compiler			
main.py	Visualize	Run ▶	Output
<pre>1 color_list1=["Red","Blue","Green","Black"] 2 print(color_list1[0:2]) #cut 1st two items 3 print(color_list1[1:2]) #cut 2nd item 4 print(color_list1[1:-2]) #cut 2nd item 5 print(color_list1[:3]) #cut 1st three items 6 print(color_list1[:]) # create copy of original list</pre>			<pre>['Red', 'Blue'] ['Blue'] ['Blue'] ['Red', 'Blue', 'Green'] ['Red', 'Blue', 'Green', 'Black'] === Code Execution Successful ===</pre>

ALGORITHMS:

ALGORITHM 1

Programiz Python Online Compiler

main.py

Visualize

Run ▶

```
1 #Algorithm-I
2 A=[1,2,3,4,5,6,7]
3 x=int(input("Enter number to find : "))
4 for i in range(len(A)):
5     if A[i]==x:
6         print("Number ",x,"found at index ",i )
7         break
8 else:
9     print("Number not found")
10
11
12
```

Output

Enter number to find : 2
Number 2 found at index 1

=== Code Execution Successful ===

main.py

Visualize

Run ▶

```
1 #Algorithm-I
2 A=[1,2,3,4,5,6,7]
3 x=int(input("Enter number to find : "))
4 for i in range(len(A)):
5     if A[i]==x:
6         print("Number ",x,"found at index ",i )
7         break
8 else:
9     print("Number not found")
10
11
12
```

Output

Enter number to find : 9
Number not found

=== Code Execution Successful ===

ALGORITHM 02

main.py

Visualize

Run ▶

```
1 A=[1,2,3,4,5,6]
2 B=[7,8,9,10,11]
3
4 x=int(input("Enter number to find : "))
5
6 found = False
7
8 for i in range(len(A)):
9     if A[i] == x:
10         print("Number", x, "found at index", i, "in Array A")
11         found = True
12         break
13
14 for i in range(len(B)):
15     if B[i] == x:
16         print("Number", x, "found at index", i, "in Array B")
17         found = True
18         break
19
20 if found == False:
21     print("Number not found")
```

Output

```
Enter number to find : 9
Number 9 found at index 2 in Array B

=== Code Execution Successful ===
```

ALGORITHM 3

main.py

Visualize

Run ▶

```
1 #Algorithm-III
2 A=[1,2,3,4,5,6]
3 B=[4,6,1,10,11]
4 found=False
5 x=int(input("Enter number to find : "))
6 for i in range(len(A)):
7     for j in range(len(B)):
8         if x==A[i] and x==B[j]:
9             found=True
10            break
11 if found:
12     print(x," is common in both arrays ")
13 else:
14     print(x," is not common in both arrays")
15
16
```

Output

```
Enter number to find : 6
6  is common in both arrays

=== Code Execution Successful ===
```

main.py

Visualize

Run ▶

```
1 #Algorithm-III
2 A=[1,2,3,4,5,6]
3 B=[4,6,1,10,11]
4 found=False
5 x=int(input("Enter number to find : "))
6 for i in range(len(A)):
7     for j in range(len(B)):
8         if x==A[i] and x==B[j]:
9             found=True
10            break
11 if found:
12     print(x," is common in both arrays ")
13 else:
14     print(x," is not common in both arrays")
15
16
```

Output

```
Enter number to find : 3
3  is not common in both arrays

=== Code Execution Successful ===
```

ALGORITHM 4

main.py

Visualize

Run ▶

Output

```
1 #Algorithm-IV
2 A=[1,2,3,4,2,1,9,8,4,5,6]
3 found=False
4 x=int(input("Enter number to chk if it is repeated in Array or not? "))
5 for i in range(len(A)):
6     for j in range(i+1,len(A)):
7         if x==A[i] and x==A[j]:
8             found=True
9             break
10 if found:
11     print(x," is duplicated in array A")
12 else:
13     print("No duplicated entity")
```

Enter number to chk if it is repeated in Array or not? 2
2 is duplicated in array A

=== Code Execution Successful ===

main.py

Visualize

Run ▶

Output

```
1 #Algorithm-IV
2 A=[1,2,3,4,2,1,9,8,4,5,6]
3 found=False
4 x=int(input("Enter number to chk if it is repeated in Array or not? "))
5 for i in range(len(A)):
6     for j in range(i+1,len(A)):
7         if x==A[i] and x==A[j]:
8             found=True
9             break
10 if found:
11     print(x," is duplicated in array A")
12 else:
13     print("No duplicated entity")
```

Enter number to chk if it is repeated in Array or not? 5
No duplicated entity

=== Code Execution Successful ===